

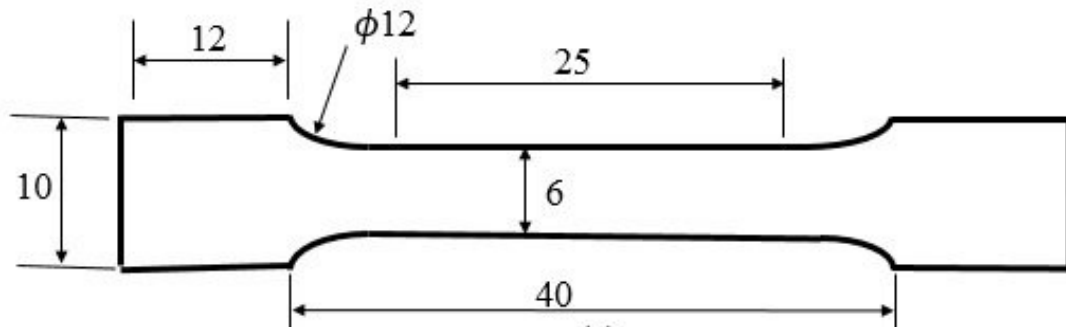
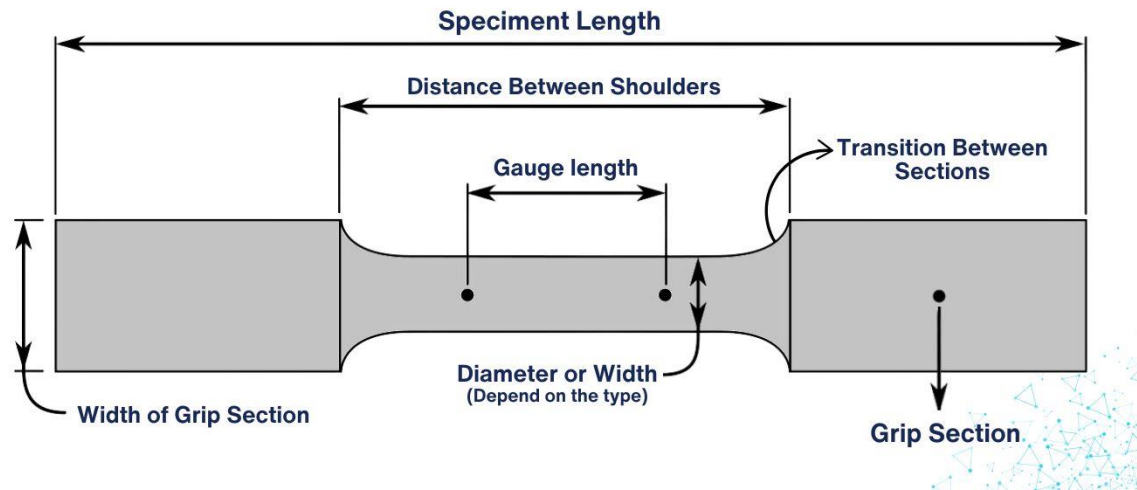
**Experiment
No. 7
Tensile
Testing**



Introduction

- **Tensile Properties:** Show how materials respond to tension forces.
- **Tensile Testing:** Specimen stretched uniaxially until fracture under a gradually increasing load.
- **Testing Equipment:** The machine elongates the specimen, measures load (load cell), and elongation (extensometer).
- **Test Output:** Records elongation vs. load, determining properties like elasticity, yield strength, and tensile strength.

Anatomy of a Tensile Test Sample



Stress-Strain Curve

Elastic Deformation: Stress and strain are proportional, following Hooke's Law.

Modulus of Elasticity (E): Proportionality constant, slope of the linear stress-strain curve.

Linear Relationship: Stress-strain curve is straight in the elastic region.

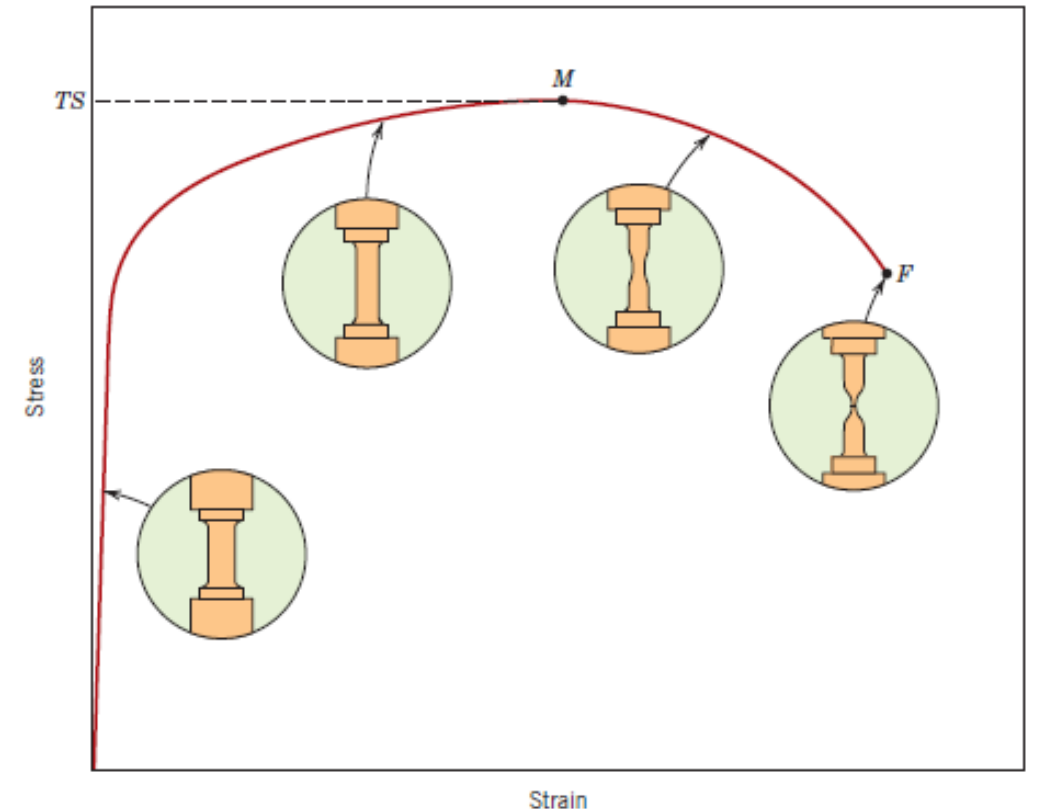
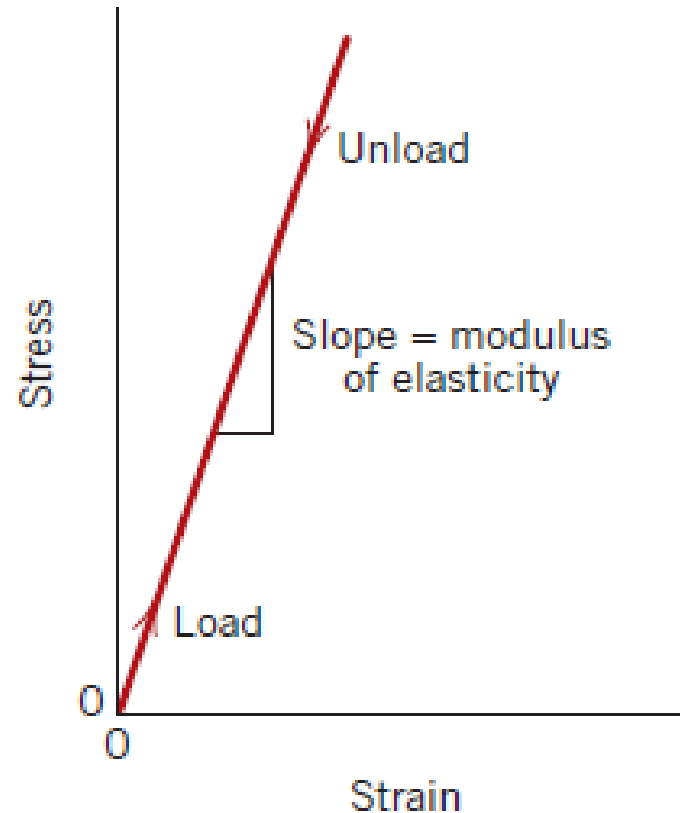
Nonpermanent: Material returns to its original shape after load removal.

Elastic Limit: Elastic deformation lasts up to a strain of ~ 0.005 .

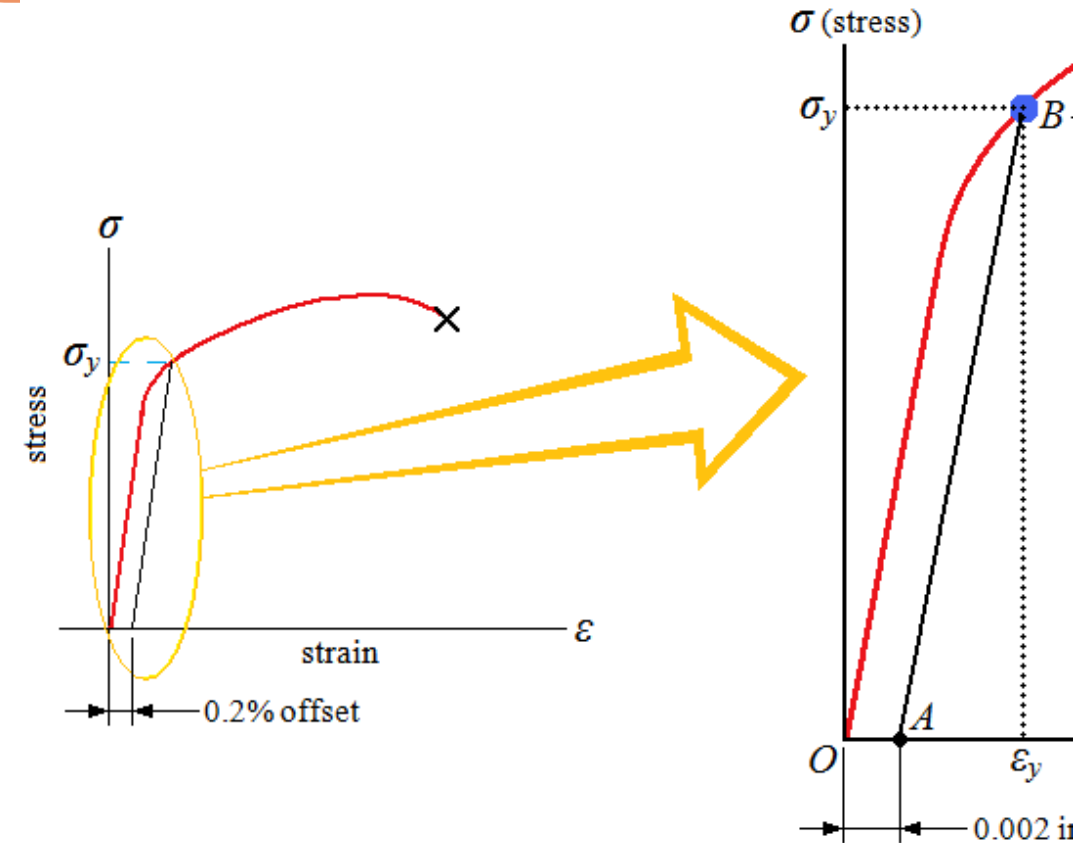
Proportional Limit: Beyond this, stress and strain are no longer proportional.

Hooke's Law: Becomes invalid past the proportional limit.

Plastic Deformation: Permanent, non-recoverable deformation occurs.



- **Yield Strength:** Stress needed for small, permanent deformation; determined using a 0.2% offset method.
- **UTS:** Maximum stress a material can endure before necking.
- **Ductility (Elongation):** Material's stretch before breaking.
- **Ductility (Reduction of Area):** Cross-sectional shrinkage at fracture.



MATERIAL DUCTILITY

is a measure of the ability of a material to deform plastically before fracture

MATERIAL TOUGHNESS

is a measure of the ability of a material to absorb energy up to fracture

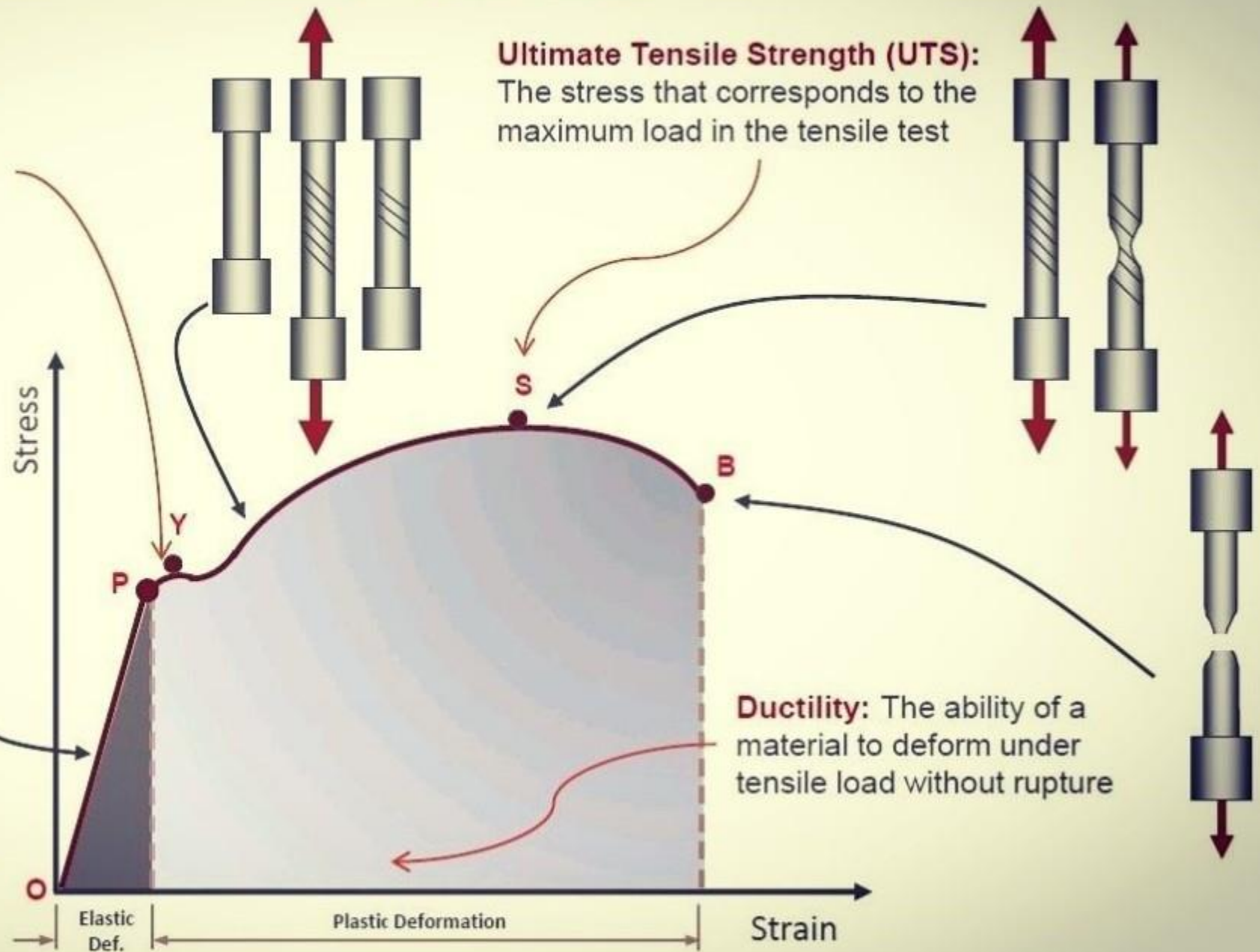
Stress – Strain Curve

Yield Strength (YS): The stress at which the plastic deformation becomes noticeable

Ultimate Tensile Strength (UTS): The stress that corresponds to the maximum load in the tensile test

Ductility: The ability of a material to deform under tensile load without rupture

P: Elastic Limit
Y: Yield Point
S: Max. Load Value
B: Breaking Point



Sample Standards

ASTM E8/E8M-11

Standard Test Methods for Tension Testing of Metallic Materials

ASTM D638 – 10

Standard Test Method for Tensile Properties of Plastics

Benefits of Tensile Testing

The data produced in a tensile test can be used in many ways including:

- To determine batch quality
- To determine consistency in manufacture
- To aid in the design process
- To reduce material costs and achieve lean manufacturing goals
- To ensure compliance with international and industry standards

Applications

- Ensures material quality across industries.
- Tests strength for aerospace and automotive parts.
- Assesses durability of packaging and construction materials.
- Verifies reliability of electronic and electrical components.
- Ensures safety of medical implants and devices.

Thank You

